



Medical Equipment Maintenance System (MEMS)



Agenda

- Problem Statement
- Key Features of MEMS
- MEMS Functional Overview
- About Us

Problems of Healthcare Organizations

No Single, Reliable Asset Register

Different departments (Biomedical, OT, ICU, Radiology, Lab, Admin, IT, Stores) hold separate records. Management often does not have a clear, consolidated view of "what we own, where it is, and in what condition".



Unplanned Breakdowns & Clinical Risk

Sudden failure of ventilators, monitors, infusion pumps, OT lights, imaging systems, or UPS can directly affect patient care and revenue.

Many breakdowns could be avoided with structured preventive maintenance.

Poor Visibility into Total Cost of Ownership

Without proper lifecycle tracking, hospitals cannot accurately see which assets are high-maintenance, which should be replaced, or how much is being spent by department or modality.

Missed Warranty / AMC / Calibration

Warranty End Dates

Tracked manually, leading to lost warranty benefits

AMC Renewals

Often overlooked, resulting in higher repair costs

Calibration Schedules

Missed timelines essential for quality and patient safety

Warranty end dates, AMC renewals and calibration schedules are often tracked manually. As a result, hospitals lose warranty benefits, pay more for repairs, or miss calibration timelines essential for quality and patient safety.



Audit & Accreditation Challenges

Pulling out equipment lists, maintenance history, calibration certificates and breakdown logs just before audits is time-consuming and error-prone.

Teams spend days compiling data instead of focusing on improvement.

 **NABH / JCI / ISO / internal audits** require comprehensive documentation



Limited Traceability & Accountability

1

Equipment Shifted

Between departments or locations
without proper documentation

2

Difficult to Track

Who is responsible for each device

3

Service History Lost

What was done and how often a
device has failed or been serviced

Equipment is shifted between departments or locations without proper documentation. It becomes difficult to track who is responsible, what was done, and how often a device has failed or been serviced.

The Cumulative Impact

Clinical Risk

Patient care directly affected by equipment failures and lack of preventive maintenance

Financial Loss

Revenue impact from downtime, missed warranties, and unnecessary repair costs

Compliance Burden

Time-consuming audit preparation and risk of accreditation issues

Operational Inefficiency

Lack of accountability, traceability, and strategic asset management

Key Departments Affected

Biomedical

Managing equipment maintenance and repairs

OT & ICU

Critical care equipment reliability

Radiology & Lab

Imaging systems and diagnostic equipment

Admin & IT

Asset tracking and documentation

Stores

Inventory and equipment distribution

Key Features of MEMS

Key Features

Centralized Asset Repository for hospital equipment

Equipment calibration tracking (NABH DOM.6 / AERB)

Automated preventive & breakdown maintenance schedule

Department-wise and floor-wise asset tracking

AMC / CMC vendor and contract management

Audit-ready document for compliance verification

Asset depreciation and financial reporting

Mobile-enabled scanning and maintenance updates



Key Features

Multilingual System

Role based Access Control(Different Menus and Screens/Role)

Extensively designed configurable workflow

Maintain different properties per Asset Types

Advanced Filter Mechanism for all Screens

Complete Audit Log of all operations within the system

View & Manage Assets based on Branches/Depts/Locations

Smart Grouping/Linkage of Assets(Child/Parent & Group Assets)



MEMS Functional Overview

Role Based Dashboard(Configurable)



Calibration due status



Preventive maintenance
compliance



Events and Notifications



Pending work orders



Asset Data



Vendor Renewals

Equipment Maintenance & Calibration Automation

- Automates periodic **maintenance** and **calibration** activities to ensure full compliance with **NABH** and **AERB** regulatory requirements
- Digitally stores and organizes **maintenance history**, **vendor service details**, and **calibration certificates** for every asset
- Provides **instant access** to records and certificates for quick **audit verification**
- Sends **intelligent alerts & notifications** to prevent missed schedules and overdue activities
- Reduces **equipment downtime** and supports **continuous compliance** with safety standards



Hospital Department-Level Asset Control

Each department (ICU, Radiology, Operation Theatre, Laboratory, etc.) has complete visibility over its assigned equipment. Department heads can view usage, maintenance history, and asset condition.





Comprehensive Asset Assignment & Tracking

Asset Assignment

Equipment assigned to departments, wards, or specific technicians

Movement Tracking

System tracks every transfer with timestamp and user identification

Audit History

Complete accountability trail maintained for regulatory compliance

Barcode Scanning

Simplified asset identification and instant status updates



Asset Serial Number Management

Unique identification ensuring precise tracking and simplified auditing

Unique Identification

Every asset assigned a unique serial number for precise tracking and audit verification

Auto-Generated Numbering

Customizable patterns for different asset types ensuring consistent identification



Serial Number Examples

- **Medical Equipment:** MED-001, MED-002
- **Furniture:** FUR-001, FUR-002
- **Laboratory Instruments:** LAB-001, LAB-002





Work Order & Vendor Management

- **Auto-generates work orders** for vendor-managed maintenance activities
- Links vendors to **specific equipment** and **AMC/CMC contracts** for clear responsibility tracking
- Tracks vendor **SLA performance**, **response times**, and service history with **management reports**
- Captures vendor **feedback against defined SLAs** to support decisions: **continue**, **replace**, or **block** a vendor for future maintenance
- Enables efficient **vendor contract management** (AMC/CMC terms, validity, renewals, documents)
- Supports **approval workflows** for vendor **onboarding** and **retention/renewal**, if required



Approval Workflow for Critical Asset Maintenance



- **Approval workflow for critical asset maintenance** to ensure control and compliance
- **Configurable multi-level approvals** involving different roles (e.g., Biomedical, HOD, Admin, Management)
- Complete **audit trail** of all approvals/rejections with **notes and timestamps** recorded in the system
- **Asset-type based rules:** different approval levels (or no approval) for different asset categories/types

Scrap and Scrap Sales – End of Life

- **Scrap assets** based on usage condition or salvage value
- Allocate and track **storage location** for scrapped items
- **Group/batch** scrapped assets and manage **scrap sales** (sell-off)
- Record complete **scrap sale details** including buyer info and **sale amount** for tracking & audit



Reports & Analytics

1

Equipment maintenance log

Complete chronological view of asset history including purchase date, commissioning details, usage patterns, and disposal information for full lifecycle transparency.

2

Calibration due list

Detailed register of all organizational assets with advanced filtering capabilities by department, vendor, assigned employee, and custom attributes.

3

Asset lifecycle summary

Comprehensive tracking of all maintenance activities, work orders, scheduled services, and vendor performance details with complete audit trails.

4

Department-wise asset cost report

Real-time display of current asset worth, depreciation calculations, book values, and total portfolio valuation for financial planning.

5

Vendor response and performance report

Captures all workflow approval steps, process timestamps, and breakdown incident details for complete operational traceability and accountability.



About Us



About Rio BizSols Pvt Ltd

Who We Are

A product-based company specializing in innovative software solutions that deliver reliable, scalable, and efficient technology to empower businesses.

Our Expertise

End-to-end software development, system integration, and business process optimization tailored to your unique operational needs.

Client Focus

We partner with organizations to streamline operations, improve productivity, and achieve meaningful digital transformation.

Our Vision: To be a trusted technology partner driving growth and innovation for enterprises worldwide.

Founder – Mr. Tony Rozario

1. 20+ years of expertise in SAP Consulting and Project Management.
2. Global experience across 9+ countries including US, Germany, Saudi Arabia & Australia.
3. Specialist in **SAP Digital Manufacturing**.
4. Skilled in **SAP UI5, HTML, JavaScript, and Integration Technologies**.
5. Deep knowledge of manufacturing processes and system automation.
6. Proven leader in managing cross-functional and global project teams.
7. Successfully delivered multiple large-scale SAP implementations.
8. Worked with leading global clients like **Saudi Aramco, Honeywell, ITC, and Tata Steel**.
9. Renowned for strong client engagement and end-to-end delivery excellence.
10. Visionary founder driving **Asset Lifecycle Management (ALM)** innovation and digital transformation.





Thank You

Ready to transform your asset lifecycle management? Let's discuss how Rio BizSols can help your organization achieve operational excellence.